

OFFICIAL TECHNICAL MANUAL

The Indian Wings Company

Comprehensive Architectural, Engineering & Production Deployment Documentation

Architecture: Next.js 16 + Express + Neon + Upstash Status: Production Ready Date: September 2026

Official Technical, Architectural & Implementation Reference Manual

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1. Executive Summary & Project Vision

1.1 Project Purpose

The Indian Wings Company is a premier luxury travel agency and destination management company specializing in the Kashmir Valley. The web platform delivers a cinematic, mobile-first booking experience tailored for high-intent domestic and international travelers seeking curated Kashmir holiday packages, luxury Dal Lake houseboats, Gulmarg ski retreats, Pahalgam alpine tours, and private chauffeured transport.

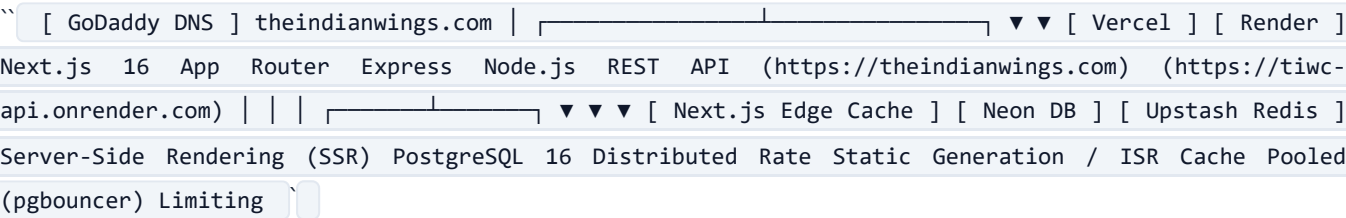
1.2 Core Architectural Principles

- **Mobile-First & High Performance:** 75%+ of travel traffic originates from mobile devices; layouts, touch targets, and image payloads are rigorously optimized for cellular speeds.

- **Zero-Loss Lead Engine:** No customer enquiry is ever lost due to network flickers, server outages, or spotty connectivity in mountain terrains. A 3-layer fallback system guarantees lead persistence.
- **Cinematic Visual Excellence:** Deep Midnight Navy #0B1F2A paired with Kashmiri Saffron #F97316 and Warm Sand #F4EFE6, enriched by smooth micro-animations and Cloudinary CDN-optimized 4K drone cinematography.
- **Complete Business Independence:** A bespoke admin control suite empowers business owners to update packages, manage hero videos, review inbound leads, inspect UTM marketing campaigns, and edit site metadata without requiring developer intervention.

2. Target Production Architecture

The system employs a decoupled, production-hardened micro-architecture:



2.1 Technology Stack Matrix

Component	Technology	Role / Environment
Frontend Framework	Next.js 16 (App Router), React 19, TypeScript	Server-rendered pages, interactive client widgets, SEO head generation
Backend API Engine	Node.js 22, Express 4.21, TypeScript	Standalone REST API with Helmet security, origin guards, and CRUD routing
Database	PostgreSQL 16 (Neon Serverless)	Authoritative ACID data store with connection pooling (pgbouncer)
Data Access Layer	Prisma ORM 7.10	Type-safe queries, migration lifecycle, and JSON column management
Distributed Cache & Rate Limiting	Upstash Redis (REST API)	Sliding-window DDoS mitigation & burst traffic defense
Media Delivery	Cloudinary CDN (next-cloudinary)	Adaptive bitrate video streaming, WebP/AVIF auto-formatting
Styling & Animation	Tailwind CSS 3.4, Framer Motion	Fluid responsive grids, bottom sheet springs, and hero fades
Validation & Security	Zod 3.25, React Hook Form	Strict schema validation, XSS input sanitization, and bot honeypot traps
Email Service	Nodemailer with SMTP	Automated itinerary delivery and internal sales lead dispatch

3. Design System & Brand Identity

3.1 Color Palette

- Midnight Navy (#0B1F2A) : Primary luxury background and high-contrast typography base. Imparts elegance, night-sky alpine depth, and premium distinction.
- Kashmiri Saffron (#F97316 / #EA580C) : High-converting accent tone used for Primary CTAs, price tags, interactive hover states, and discount badges.
- Warm Sand (#F4EFE6) : Soft neutral foundation for background contrast, card backgrounds, and pill borders.

- **Forest Alpine Emerald (#064E3B / #059669)** : Trust indicators, WhatsApp actions, verified reviews, and seasonal badge highlights.

3.2 Typography Hierarchy

- **Headlines:** *Playfair Display* (Serif) – Evokes heritage, bespoke hospitality, and luxury vacation aesthetics.
- **Body & Controls:** *Manrope* (Sans-Serif) – Highly legible, contemporary geometric sans-serif optimized for mobile screens and data density.

3.3 UI Components & Micro-Interactions

- **Glassmorphic App Bars:** Floating backdrop-filter navigation with dynamic scroll-depth tinting.
- **MakeMyTrip (MMT) Style Bottom Sheets:** Spring-animated drawer sheets for zero-friction mobile lead submissions.
- **Interactive Badges:** Real-time status indicators ("Instant Confirmation", "Best Seller", "Snow Season Special").

4. Data Architecture & Schema Specifications

The PostgreSQL database is organized into 11 interconnected relational models:

```
mermaid erDiagram
    Enquiry ||--o| UserSession : "linked by visitorId"
    UserSession ||--o{ SessionEvent : "contains"
    PackageCategory ||--o{ Package : "categorizes"
    Package {
        string id PK
        string slug UK
        string title
        string duration
        int startingPrice
        json itinerary
        boolean isActive
    }
    Enquiry {
        string id PK
        string name
        string phone
        string travelDate
        string status
        string utmSource
    }
    UserSession {
        string id PK
        string visitorId
        boolean converted
        int durationSeconds
    }
    Destination {
        string id PK
        string slug UK
        string name
        string elevation
        string bestSeason
    }
    Activity {
        string id PK
        string slug UK
        string name
        string duration
        int priceFrom
    }
    TransportVehicle {
        string id PK
        string slug UK
        string name
        string seats
        int pricePerDay
    }
    TransportRoute {
        string id PK
        string routeTitle
        string distance
        int startingPrice
    }
    CustomerReview {
        string id PK
        string name
        string city
        int rating
        string review
    }
    SiteSetting {
        string id PK
        string siteTitle
        string phone
        string whatsapp
        json heroConfig
    }
```

4.1 Schema Models Breakdown

1. **Enquiry** : Records all customer booking inquiries with UTM parameters (utmSource , gclid , fbclid), lead status lifecycle (NEW , CONTACTED , QUOTED , WON , LOST), and IP address.
2. **UserSession** : Captures visitor browser journeys, device types, screen resolution, referrers, and conversion flags.
3. **SessionEvent** : Micro-event tracking within sessions (scroll depth %, card clicks, modal triggers, dwell time).
4. **Package & PackageCategory** : Rich travel package

listings with comprehensive day-by-day JSON itineraries, inclusions, exclusions, and SEO tags. 5.

Destination : Kashmir valley hubs (Gulmarg, Pahalgam, Sonamarg, Dal Lake, Doodhpathri, Yusmarg)

with elevation, distance from Srinagar, and seasonal guides. 6. **Activity** : Winter skiing, Dal Lake shikara rides, paragliding, river rafting, and pony trekking. 7. **TransportVehicle** &

TransportRoute : Chauffeur fleet (Innova Crysta, Urbania, Tempo Traveller, 4x4 Thar) and intercity transfer routes. 8. **CustomerReview** : Verified traveler ratings, written testimonials, and video review links. 9. **SiteSetting** : Global site titles, announcements, phone/WhatsApp numbers, custom robots.txt, and homepage hero configuration.

5. Lead Engine & Zero-Loss Resilience Pipeline

To eliminate lost revenue from mobile network dropouts, the platform implements a **Three-Layer Zero-Loss Pipeline**:

[User Submits Enquiry] | ▼ [Layer 1: Client-Side Offline Queue (IndexedDB)] Stores payload locally before dispatch. If device loses internet, retries automatically on reconnect. | ▼ [Layer 2: Secure Express / Next.js REST API]

- Bot Trap: Hidden honeypot field inspection.
- Upstash Rate Limit: 5 requests / 10 minutes.
- Idempotency: Duplicate check within 5 minutes.
- Atomic DB Write: Persisted into PostgreSQL.
- Email Service: Sends itinerary & alerts sales desk.

| (If Server Errors / 500 Failure) ▼ [Layer 3: Instant WhatsApp Fallback (wa.me)] Redirects customer to pre-filled WhatsApp chat with Srinagar desk specialists. Zero lead attrition under any conditions.

6. MakeMyTrip (MMT) UI Placement & Conversion Blueprint

Drawing inspiration from India's highest-converting travel applications, conversion mechanisms are embedded at critical points:

1. **Sticky Bottom Mobile Bar**: Persistent bottom action drawer on mobile viewports featuring "Get Instant Quote" and direct WhatsApp access.
2. **Inline Contextual Triggers**: Dynamic package cards pass the package title, duration, and price directly into the enquiry modal.
3. **Instant Itinerary Gate**: Customers can download comprehensive PDF-formatted itineraries by entering their phone/email, capturing high-intent leads earlier in the research phase.

7. Admin Portal, Content Management & Live Telemetry

An isolated administration console is accessible at `/admin` :

Admin Portal Dashboard |— Overview & Metrics (Total Leads, Leads Today, Conversion %, Average Dwell Time) |— Inbound Leads Inbox (Status workflow: NEW → CONTACTED → WON) |— Marketing Campaigns (UTM clicks, conversion rates per Google/Meta campaign) |— Homepage Hero Editor (Change headline, badge, video URL, slides & trust pills) |— Tour Packages Manager (Full CRUD with visual day-by-day itinerary builder) |— Destinations & Activities (Add seasonal spots, change pricing & photos) |— Transport Fleet & Routes (Manage vehicle fleet and route rates) |— Customer Reviews (Manage written testimonials and video reviews) |— SEO & Settings (Edit site title, meta description, contact info & robots.txt)

7.1 Security Architecture

- **Cryptographic Session Cookies:** HMAC-SHA256 authenticated `tiwc_admin_session` cookie.
- **Cross-Origin Compatibility:** Set with `SameSite=None; Secure` so the Vercel-hosted admin interface securely queries the Render backend API.
- **Timing-Safe Password Verification:** Protects against side-channel timing attacks.
- **Distributed Rate Limiting:** Upstash Redis throttles failed login attempts (5 attempts per 15 minutes).

8. Google SEO Engine, Structured Data & Sitemaps

8.1 On-Page Optimization

- **Single `<h1>` Hierarchy** : Validated semantic HTML on all landing pages and dynamic package routes.
- **OpenGraph & Twitter Cards:** High-res Cloudinary preview banners with optimized titles and descriptions.

- **Dynamic sitemap.xml** : Automatically pulls active packages and destinations from the database, updating in real time.
- **Dynamic robots.txt** : Served directly from database `SiteSetting.robotsTxtCustom` with sensible crawl fallbacks.

8.2 JSON-LD Structured Data

All package detail routes output Google-compliant `Product` and `TouristTrip` structured schemas:

```
`json { "@context": "https://schema.org", "@type": "TouristTrip", "name": "Kashmir Honeymoon Special", "description": "Romantic 6-day Kashmir package with private houseboat stay.", "offers": { "@type": "Offer", "price": "24999", "priceCurrency": "INR", "availability": "https://schema.org/InStock" } } `
```

9. Media & Cloud Delivery Specifications

9.1 Video & Photography Optimization

- All visual assets are stored on **Cloudinary CDN**.
- Desktop hero background uses high-bitrate MP4 with `f_auto,q_auto,ac_none` transformations.
- Images utilize dynamic Next.js image loaders, delivering WebP format on Chrome/Android and AVIF on iOS/macOS.
- First contentful paint (FCP) is prioritized with preloaded poster stills (`so_0,f_auto,q_auto`).

10. Third-Party Services Directory

Provider	Purpose	Environment Variable	Criticality
Neon	Serverless PostgreSQL Database	DATABASE_URL	Critical
Upstash	Distributed Redis Rate Limiting	UPSTASH_REDIS_REST_URL , ..._TOKEN	High
Render	Node.js Express REST API Hosting	PORT , NODE_ENV	Critical
Vercel	Next.js Frontend App Hosting	NEXT_PUBLIC_SITE_URL , NEXT_PUBLIC_BACKEND_URL	Critical
Cloudinary	Global Media CDN & Video Engine	Asset URLs (pre-configured)	High
Microsoft Clarity	Session Recordings & Heatmaps	NEXT_PUBLIC_CLARITY_ID	Optional
Google Analytics 4	Visitor Funnel & Ad Tracking	NEXT_PUBLIC_GA_MEASUREMENT_ID	Optional
GoDaddy	Primary Domain Registration & DNS	A & CNAME records	Critical

11. Complete Implementation Roadmap (Phases 1–15)

- **Phase 01:** Project foundation, Tailwind design system, and TypeScript strict setup.
- **Phase 02–02C:** Glassmorphic Navbar, video hero with poster fallbacks, and inline lead forms.
- **Phase 03–03D:** Trust signals ("Why Travel With Us"), destination detail pages, and itinerary guides.
- **Phase 04:** Tour packages listing engine with category filtering (Featured, Honeymoon, Family).
- **Phase 07A:** Global floating quick-action contact widgets (WhatsApp & Phone).
- **Phase 08:** Trusted hotel partners and Kashmir tourism collaborator marquee.
- **Phase 09:** PostgreSQL database integration, Prisma ORM, and Zero-Loss Offline Queue.
- **Phase 10:** MakeMyTrip (MMT) mobile bottom drawer sheet and dynamic package preselection.
- **Phase 11:** Isolated Admin Portal with visitor activity telemetry and UTM campaign analytics.
- **Phase 12:** Google SEO rich snippets, package schema generators, and admin sitemap controls.
- **Phase 13:** Separate backend extraction to standalone Express REST API on Render.
- **Phase 14:** Distributed Upstash Redis rate limiting and cross-origin admin auth.
- **Phase 15:** Production readiness audit, security header verification, and final documentation.

12. Production Deployment & Infrastructure Runbook

12.1 Render Backend Setup

1. Create a **Web Service** on [Render](https://render.com) connecting the repository. 2. Set **Root Directory** to `backend`. 3. Set **Build Command**: `npm install && npx prisma generate && npm run build`. 4. Set **Start Command**: `npm start`. 5. Configure environment variables from `backend/.env.example`. 6. Verify health probe: `curl https://<your-render-api>/health` returns `{"status":"ok","db":"connected"}`.

12.2 Vercel Frontend Setup

1. Connect repository on [Vercel](https://vercel.com). 2. Set Framework Preset: **Next.js**. 3. Set `NEXT_PUBLIC_BACKEND_URL` to your Render API endpoint. 4. Set `DATABASE_URL` (for server-side rendering fallback). 5. Deploy.

12.3 GoDaddy DNS Records

- `@` (Apex A Record) → 76.76.21.21 (Vercel IP)
- `www` (CNAME Record) → `cname.vercel-dns.com`
- `api`` (Optional CNAME) → Point to Render custom domain target

Document compiled and verified for The Indian Wings Company production launch.
